

```

import numpy as np

def incmatrix(genl1,genl2):
    m = len(genl1)
    n = len(genl2)
    M = None #to become the incidence matrix
    VT = np.zeros((n*m,1), int) #dummy variable

    #compute the bitwise xor matrix
    M1 = bitxormatrix(genl1)
    M2 = np.triu(bitxormatrix(genl2),1)

    for i in range(m-1):
        for j in range(i+1, m):
            [r,c] = np.where(M2 == M1[i,j])
            for k in range(len(r)):
                VT[(i)*n + r[k]] = 1;
                VT[(i)*n + c[k]] = 1;
                VT[(j)*n + r[k]] = 1;
                VT[(j)*n + c[k]] = 1;

            if M is None:
                M = np.copy(VT)
            else:
                M = np.concatenate((M, VT), 1)

            VT = np.zeros((n*m,1), int)

    return M

```

Listing 1: Example from external file

```

#include <stdio.h>
int main() {
    printf("Hello, World!"); /*printf() outputs the quoted string
    return 0;
}

```

Listing 2: Hello World in C

minted makes a nice job of typesetting listings `??`, `??` and `??`.

List of source codes

```
function fact (n)--defines a factorial function
if n == 0 then
return 1
else
return n * fact(n-1)
end
end

print("enter a number:")
a = io.read("*number") -- read a number
print(fact(a))
```

Listing 3: Example from the Lua manual